

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 35%

QUESTIONS: 5

Duration : 1 Hour
Total Marks:50

Annexure A

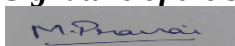
Analytical Problem Solving

Code of Conduct:

I M.Pranai(192525068) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

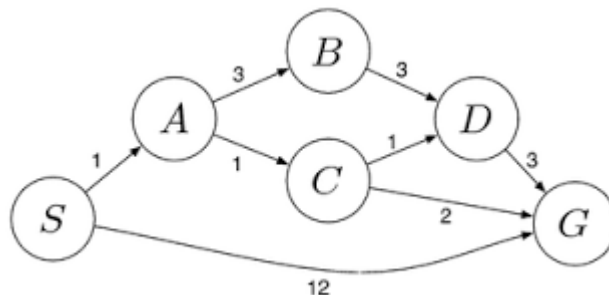


Annexure A

Analytical Problem Solving

1. Solve the Water Jug problem: you are given 2 jugs, a 4-gallon one and 3-gallon one. Neither has any measuring maker on it. There is a pump that can be used to fill the jugs with water. How can you get exactly 2 gallons of water into 4-gallon jug? Explicit assumptions: A jug can be filled from the pump, water can be poured out of a jug onto the ground, water can be poured from one jug to another and that there are no other measuring devices available.
2. Find out how Mars Rover, analyse and explore the samples on Mars by answering the following questions.
 - i. What are the percept's for this agent?
 - ii. Characterize the operating environment.
 - iii. What are the actions the agent can take?
 - iv. How can one evaluate the performance of the agent?
 - v. What sort of agent architecture do you think is most suitable for this agent? Why?
3. Explore the search space using search strategies and formulate the problem components for the 8-Queens problem using the following information: Place 8 queens on a chessboard such that none of the queen's attack any of the others. A configuration of 8 queens on the board as shown in the figure below, but this does not represent a solution as the queen in the first column is on the same diagonal as the queen in the last column.
4. A customer wants to travel from the one location to another location using OLA Cab booking mobile application. The customer can select the any of the cab type as mini, micro, sedan, shared, prime and so on based on his comfort to reach the destination. Identify what type of problem-solving agent can be used and write the pseudocode for the above problem.
5. A logistics company operates a delivery network where packages must be transported between warehouses at minimum cost. Each route between warehouses has an associated travel cost (fuel + time). The company wants to determine the least-cost path from the starting warehouse S to the destination warehouse G using the Uniform Cost Search (UCS) algorithm.

The delivery network is represented as a weighted graph:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem
Method Selection	20%	Selects most appropriate method/formula with strong justification	Selects correct method with minor errors	Inappropriate or partially correct method	Unable to select method
Solution Process	25%	Logical, systematic steps with correct approach throughout	Mostly correct steps with minor mistakes	Disorganized steps; multiple errors	No clear process
Accuracy of Calculation	20%	All calculations accurate; correct final answer	Minor calculation errors	Frequent errors; incorrect final answer	No valid calculations
Interpretation of Results	15%	Clearly interprets results with units and engineering relevance	Basic interpretation with minor omissions	Limited or unclear interpretation	No interpretation
Problem Understanding	20%	Clearly interprets problem, identifies all parameters and requirements accurately	Understands problem with minor gaps	Partial understanding; misses key elements	Misinterprets or unable to understand problem

ANSWER

1. Water Jug Problem

Problem Understanding

We have two jugs of capacity **4 gallons and 3 gallons**. There are no measuring marks. Using only filling, emptying, and pouring operations, we have to obtain exactly **2 gallons in the 4-gallon jug**.

State Representation

Let the state be **(x, y)**, where:

- x = water in 4-gallon jug
- y = water in 3-gallon jug
- Initial state = **(0,0)**
- Goal state = **(2,0)**

Process:

Step	Operation	Explanation	State
0	Start	Both jugs are empty	(0,0)
1	Fill 3-gallon	Fill the 3-gallon jug completely	(0,3)
2	Pour 3 → 4	Transfer all 3 gallons into 4-gallon jug	(3,0)
3	Fill 3-gallon	Fill the 3-gallon jug again	(3,3)
4	Pour 3 → 4	Only 1 gallon can enter the 4-gallon jug; 2 remain	(4,2)
5	Empty 4-gallon	Empty the 4-gallon jug	(0,2)
6	Pour 3 → 4	Transfer the remaining 2 gallons	(2,0)

Final Result

The **4-gallon jug contains exactly 2 gallons**. The goal is achieved in **6 moves**.

2. Mars Rover – PEAS Analysis

Introduction

The Mars Rover is an autonomous intelligent agent that explores the Martian surface, collects samples and analyses them. It receives information from its surroundings and chooses suitable actions.

i. Percepts

Percepts are the information received by the rover from its environment.

Percept	Information
Camera	Images of rocks, terrain and obstacles
Motion sensors	Position, movement and inclination
Sample instruments	Soil and rock information
Environment sensors	Temperature, dust and radiation
Power information	Battery/solar power level
Communication	Commands received from Earth

ii. Operating Environment

Property	Characterization	Explanation
Observability	Partially observable	Rover can observe only its surroundings
Agents	Single agent	Rover operates independently
Determinism	Stochastic	Terrain and wheel movement can be uncertain
Sequential	Sequential	Current actions affect future actions
Dynamic	Dynamic	Weather, dust and lighting can change
Continuous	Continuous	Position, speed and time vary continuously
Known	Partly unknown	Rover may encounter unexplored terrain

iii. Actions

The rover can:

1. Move forward and backward.
2. Turn and steer.
3. Stop or avoid obstacles.
4. Collect soil and rock samples.
5. Analyse samples.
6. Capture images.
7. Send information to Earth.
8. Manage its power.

iv. Performance Measure

The rover's performance can be evaluated using:

- Number and scientific value of samples collected.
- Area explored safely.
- Ability to avoid obstacles and damage.
- Efficient use of energy.
- Quality of information transmitted to Earth.
- Successful completion of exploration tasks.

v. Suitable Agent Architecture

Utility-based agent with learning capability is suitable.

Reasons

- The environment is partially observable.
- Communication with Earth has a delay, so the rover must make decisions independently.
- The rover has several goals such as exploration, safety, sample collection and energy conservation.
- A utility function helps choose the best action among different alternatives.
- Learning capability can improve navigation and terrain identification.

Conclusion

Therefore, a utility-based agent with learning capability and reactive behaviour is suitable for the Mars Rover

3. 8-Queens Problem

Problem Understanding

The objective is to place 8 queens on an 8×8 chessboard so that no two queens attack each other.

Two queens must not be in:

- Same row
- Same column
- Same diagonal

Component	Description
Initial State	Empty chessboard
State	Arrangement of queens on the board
Actions	Place a queen in a safe square
Transition Model	Add the queen to the selected square
Goal Test	8 queens placed with no attacks
Path Cost	Each placement can have cost 1

Step-by-Step Search Process

1. Start with an empty chessboard.
2. Place a queen in the first column.
3. Check whether the position is safe.
4. Move to the next column.
5. Place another queen in a non-attacking position.
6. Continue column by column.
7. If a column has no safe position, go back to the previous column.
8. Move the previous queen to another safe position.
9. Continue until all 8 queens are placed.
10. Check that no pair of queens attacks each other.

Search Strategy: Backtracking

Flow:

Start → Place Queen → Check Safety → Safe → Next Column

If unsafe:

Conflict → Backtrack → Try Another Position

Valid Solution

A valid arrangement is:

[1, 5, 8, 6, 3, 7, 2, 4]

Here each number represents the row position of the queen in columns 1 to 8.

Conclusion

The problem can be efficiently solved using **Depth-First Search with Backtracking**, because unsuccessful arrangements can be abandoned immediately.

4. OLA Cab Booking – Problem-Solving Agent

Problem Understanding

A customer wants to travel from a pickup location to a destination using the OLA application.

The customer can choose different cab types such as Mini, Micro, Sedan, Shared and Prime.

Type of Agent

Goal-based agent with utility-based selection.

The main goal is to reach the destination, while utility can be used to select the cab based on:

- Fare
- Comfort
- Waiting time
- Travel time
- User preference

Problem Formulation

Component	Description
Initial State	Customer is at pickup location
Goal	Reach destination
Actions	Enter locations, select cab, book, travel
State	Location, available cabs, fare and travel time
Performance	Low cost, less waiting, comfort and safety

Process

Step	Action
1	Enter pickup location
2	Enter destination
3	Find available cabs
4	Display cab types
5	Calculate fare and estimated arrival
6	Compare available cabs
7	Select suitable cab
8	Confirm booking
9	Track cab
10	Start journey
11	Reach destination
12	Complete payment

Conclusion

The OLA system can use a goal-based agent to achieve the destination goal and a utility-based mechanism to select the most suitable cab.

5. Uniform Cost Search – UCS

Problem Understanding

Route	Cost
S → A	1
S → G	12
A → B	3
A → C	1

Route	Cost
B → D	3
C → D	1
C → G	2
D → G	3

UCS Process

Step	Expanded Node	Calculation	Frontier
1	S(0)	A=1, G=12	A:1, G:12
2	A(1)	B=1+3=4, C=1+1=2	C:2, B:4, G:12
3	C(2)	D=2+1=3, G=2+2=4	D:3, B:4, G:4
4	D(3)	G=3+3=6, which is worse than 4	B:4, G:4
5	B(4)	D=4+3=7, which is worse than 3	G:4
6	G(4)	Goal reached	—

Path Calculation

The selected path is:

$S \rightarrow A \rightarrow C \rightarrow G$

Total cost:

$S \rightarrow A = 1$

$A \rightarrow C = 1$

$C \rightarrow G = 2$

Therefore,

Total Cost = $1 + 1 + 2 = 4$

Comparison

Path	Total Cost	Result
$S \rightarrow A \rightarrow C \rightarrow G$	4	✓ Optimal
$S \rightarrow A \rightarrow C \rightarrow D \rightarrow G$	6	Not optimal

Path	Total Cost	Result
$S \rightarrow A \rightarrow B \rightarrow D \rightarrow G$	10	Not optimal
$S \rightarrow G$	12	Not optimal

Final Result

Least-cost path = $S \rightarrow A \rightarrow C \rightarrow G$

Minimum cost = 4